PROJECT REPORT

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BRANCH : B-TECH CSE(AI-ML)

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Code Explainer AI using Generative AI

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1. Introduction

Understanding programming code can be challenging for beginners, especially when they encounter unfamiliar syntax or complex logic. Traditional learning resources often require significant time and effort to interpret code effectively.

This project, *Code Explainer AI*, aims to simplify code understanding by using Generative Artificial Intelligence. It provides clear and beginner-friendly explanations of code snippets, making programming more accessible and easier to learn.

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2. Problem Statement

Many students and beginner programmers face difficulty in understanding code due to:

- Complex syntax and logic
- Lack of proper explanations
- Time-consuming debugging and learning process

There is a need for a tool that can instantly explain code in a simple and understandable manner.

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3. Objective

The main objectives of this project are:

- To develop a web-based application that explains code
- To simplify programming concepts for beginners
- To provide structured explanations including:
    - Code functionality
    - Line-by-line explanation
    - Simple summary

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4. Proposed Solution

The proposed solution is a web application where users can:

1. Paste any code snippet
2. Click on the “Explain Code” button
3. Receive a detailed explanation generated by AI

The system uses Generative AI through the **Groq API** to process the input and generate human-readable explanations.

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5. Methodology

The working of the system follows these steps:

1. The user enters code into the web interface

2. The frontend sends the code to the backend using HTTP requests
3. The backend processes the input and sends it to the AI model
4. The AI model generates an explanation
5. The explanation is returned and displayed on the webpage

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6. Technologies Used

- **Python** – Backend programming language
- **Flask** – Web framework for building the application
- **Groq API** – For generating AI-based explanations
- **HTML & CSS** – Frontend design
- **JavaScript (Fetch API)** – For frontend-backend communication

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7. System Architecture

The system follows a simple client-server architecture:

- **Frontend (Client):**
  - User interface for input and output
- **Backend (Server):**
  - Flask application handling requests
- **AI API:**
  - Processes code and returns explanations

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8. Features

- User-friendly web interface
- Supports multi-line code input
- Generates structured explanations:
  - What the code does
  - Line-by-line explanation
  - Simple summary
- Fast and efficient responses

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9. Challenges Faced

During the development of the project, the following challenges were encountered:

- Difficulty in API integration
- Model selection issues
- Handling errors and server crashes
- Ensuring proper frontend-backend communication

These challenges were resolved through debugging and testing.

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10. Results

The project successfully:

- Accepts user input code
- Processes it using AI

- Displays clear and understandable explanations

The application works efficiently and meets the project objectives.

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11. Learning Outcomes

Through this project, the following concepts were learned:

- API integration in Python
- Working with Generative AI models
- Flask web development
- Frontend-backend communication
- Debugging and error handling

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12. Future Scope

The project can be improved further by:

- Adding support for multiple programming languages
- Enhancing UI/UX design
- Adding syntax highlighting
- Deploying the application online
- Supporting voice-based input

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13. Conclusion

The *Code Explainer AI* project demonstrates how Generative AI can be used to simplify programming education. It helps beginners understand code more effectively and reduces the learning curve.

This project highlights the practical application of AI in real-world problem-solving.